Comparing a Formula-Built and a Search-Based Quantum Compiler through a Trained Noisy Classifier: What the Comparison Must Control For

TN Holdings
love.os.architect@proton.me
https://github.com/TN-Holdings-LLC/psf-zero

Version 1 — 23 October 2026

Abstract

PSF-Zero places a quantum classifier's entangling ring on hardware by formula, without a layout search and without reading the day's calibration; Qiskit's preset pass manager searches, using the calibration. We ask which of the two gives the better trained classifier, and report what it took to answer that question honestly. The model is a 10-qubit, 16-layer data re-uploading classifier trained by a finite-difference optimiser on a noise model of `ibm_marrakesh` (the *twin*) and tested on an independently perturbed noise model (the *stand-in*); one day of hardware runs is included. Four effects can each produce a winner without any difference in learning: (i) the task's answer sat at the untrained starting point — on the device day's task the untrained model is already 0.934 accurate (noiseless), and training lowered both compilers' models to 0.762 (PSF-Zero) and 0.730 (Qiskit); (ii) a readout offset makes a circuit answer one class on 84–94 % of balanced test points, which moves accuracy while leaving the model unchanged; (iii) arms with different output scales lose different amounts to a fixed shot count; (iv) the twin's scale was off by up to 2.3× on hardware. With the task moved far from the start and the rules fixed before each run, PSF-Zero's newest ring schedule (the *zipper*, 336 CZ and 11.7 μs at 16 layers) and Qiskit (506 CZ, 48.2 μs, 14 active qubits) gave classifiers that the declared tests could not tell apart: on stand-in accuracy the zipper won on one of three teachers at each budget, below the declared two of three, and on the trained model itself neither arm won on any teacher. Formula placement could not be told apart from the search while using a third fewer entangling gates and a quarter of the duration; it did not beat it. We give the vanity and the essential number side by side throughout, mark what was declared before the data and what was found after, and close with a checklist for compiler comparisons through trained quantum models.

1 Introduction

A compiler for near-term quantum hardware is usually judged on circuit metrics: two-qubit gate count, depth, duration, estimated success probability. The users of a variational quantum classifier care about something else: whether the trained model is better. Comparing compilers through a trained model is tempting, because it measures what users want, and dangerous, because training, readout and finite sampling each add their own ways for one arm to win.

The question behind this study is the design question of PSF-Zero [1]. Its AI front end places the classifier's ring of CZ gates on a heavy-hex line by a closed formula and routes it with a fixed schedule; it does not search over layouts and does not read the day's calibration. Qiskit [2] at optimisation level 2 with the device's `Target` searches for a layout and routing, and its choices use

the calibrated error rates. A formula gives a shorter, reproducible circuit; a search can pick better qubits. Which gives the better classifier?

This paper reports that comparison and, at equal length, the four effects that made earlier answers to it unreliable:

1. **A task whose answer is at the start.** Teacher–student tasks with small teacher angles put the teacher near the student's initial parameters $\theta = 0$. The untrained model is then already accurate, and training mostly moves it away (Section 4).

2. **A readout offset.** Accuracy is the sign of the readout. A circuit whose noise shifts the readout answers one class far more often than the label prior, and its accuracy changes without any change in the model (Section 5).

3. **Unequal shot fragility.** An arm whose output is shrunk more by noise needs more shots for the same decision; at a fixed shot count it loses more (Section 6).

4. **A device model with the right shape and the wrong scale** (Section 3).

We then report three declared tests on teachers placed far from the start (Sections 7.2–7.3). The principle throughout: the headline ("vanity") number is given next to the number that survives the checks ("essential"), and every result says whether its rule was fixed before the run.

Related work. Noise flattens training landscapes [4] and, when global and depolarising, shrinks expectation values uniformly [5, 6]. Nguyen et al. [7] separate the non-unital bias of amplitude damping from its contraction in variational circuits and find that a trainable output scale recovers most of the accuracy difference, which then reappears as a shot cost; our offset and shot findings are the readout-side counterparts of that picture for whole compiled circuits. Kashif and Shafique show that transpilation changes gradient variance and expressibility compared with the logical circuit, in noiseless simulation and without training to an accuracy [8, 9]; we compare two exact compilations under noise through the trained model. Bowles, Ahmed and Schuld show how much benchmark design decides quantum machine learning comparisons [10]; the confounds here are specific to noisy readout and are not covered there. Swap networks route all-to-all or ring interactions on a line [11, 12]; the zipper's merge of a CZ with the following SWAP into a two-CZ block is standard gate algebra, and we claim no novelty for it.

2 System

2.1 Model and task

The classifier is a data re-uploading model [3] on $n = 10$ qubits with $L = 16$ layers. Each layer applies $R_Y(\frac{\pi}{2}x_q)$ to every qubit q, then trainable $R_Y(a_q)R_Z(b_q)$, then CZ on the ring $(0,1),\ldots,(n-1,0)$; a final trainable $R_Y(c_q)$ closes the circuit, and the prediction is the sign of $z = \langle Z_0 \rangle$. There are $2nL + n = 330$ parameters.

The task is teacher–student. A teacher of the same architecture draws its angles uniformly from $[-a, a]$; inputs are uniform in $[-1, 1]^{10}$; the label is the sign of the teacher's z, and points with $|z| < 0.2$ are dropped. Classes are balanced: 64 training and 500 test points, half of each sign, unless stated otherwise. With $a = 0.3$ (all earlier work, including the device day) the teacher is near $\theta = 0$; the far-teacher tests use $a = 1.0$.

2.2 Three compilations of the same logical circuit

	item 31 (PSF-Zero)	zipper (PSF-Zero)	Qiskit 2.5.2, level 2
placement	formula (ring on a line)	formula (ring on a line)	search, calibration-aware
routing per layer	6 plain SWAPs	merged CZ+SWAP blocks	search
CZ per layer	28	$21 = 5n/2 - 4$	–
CZ, $L = 16$	448	336	506
duration, $L = 16$	18.0 μs	11.7 μs	48.2 μs
active qubits	10	10	14

Table 1: The three arms on the `ibm_marrakesh` snapshot of 2026-10-13. Durations are the scheduled circuit durations on that Target; CZ counts are from the compiled circuits.

Item 31 is PSF-Zero's ring template as used on the device day: the ring is laid on a line of 10 physical qubits and its far edges are brought together by plain SWAPs (3 CZ each). The *zipper* lays the ring out so that pairs alternate between its two arcs; the two end SWAPs of each layer then fall on ring edges and merge with their CZ into one two-CZ block, and odd layers run the rounds backwards to undo the permutation. It costs $5n/2 - 4$ CZ per layer. An exhaustive search over schedules starting from minimal-length layouts found no cheaper schedule for $n = 4, 6, 8$; for $n = 10$ the search did not finish and 21 is not proven minimal. Both PSF-Zero arms are built by formula, without search; the only device data they read are the gate durations. On the device day item 31 was built by PSF-Zero's Python front end (a22); in the far-teacher tests both arms were built by a24's Rust ring builder (`psfbind` ring r2), checked identical to a22's output at build time. The Qiskit arm is TOQB's reference: the parameterised circuit transpiled once at level 2 with the device's `Target` and bound per evaluation (`qiskit:2:target:template`).

All three compiled circuits implement the logical model exactly: with every noise source switched off, the twin's fitted scale is 1.000 ($|1 - \lambda| < 1.1 \times 10^{-5}$) for both PSF-Zero arms at $L = 8$ and 16, and the zipper's statevector matches the logical circuit to 4.7×10^{-16}.

2.3 Evaluation tiers

- **Noiseless:** the logical model's exact z. Used to read the trained model itself.
- **Twin:** TOQB's stochastic-trajectory noise model of the `ibm_marrakesh` calibration snapshot of 2026-10-13 (depolarising gate errors composed with T_1/T_2 relaxation over each gate, relaxation over idle delays after ALAP scheduling, readout error). Training uses it exclusively.
- **Stand-in:** the same snapshot with every qubit's T_1 and T_2 scaled by factors drawn once in $[0.7, 1.3]$, a systematic 1 % over-rotation of every `sx` and `x`, and a static 10 kHz ZZ coupling between neighbours accumulated over the schedule, idle time included. It is evaluated by 1,024 stochastic trajectories per test point (the trajectory engine is checked against the exact density matrix). All test results below are on the stand-in unless marked otherwise. Its $P(1)$ per test point is recorded, so accuracy can be read without shot noise ("shot-free") or with the run's 250 shots.
- **Device:** one job on `ibm_marrakesh` on 2026-10-22 (Section 3).

The stand-in shares the twin's calibration base. It is a test of the twin's robustness to plausible model errors, not an independent device, and agreement between them is optimistic.

2.4 Training and statistics

Training ("design C") runs entirely on the twin: from $\theta = 0$, each loop estimates the gradient of the loss $\frac{1}{4}\sum_b(1 - y_b z_b)/2$ on a batch of 4 training points by antithetic finite differences along 8 random ± 1 directions with step $h = 0.1$ (32 trajectories, common random numbers), and moves θ by learning rate 0.4; seed 301. The trained model is then predicted by the twin and verified once on the stand-in.

Paired comparisons on the same 500 test points use the exact McNemar test; Wilcoxon signed-rank tests are used for per-point differences; Holm's correction is applied over each rule's comparisons at $\alpha = 0.01$. A rule's verdict needs at least 2 of 3 teachers. AUC (the ranking of points by the output, offset- and scale-free) is reported as a descriptive diagnostic.

3 The device day: right shape, wrong scale, and a task that needed no learning

On 2026-10-22 one job ran item 31 and Qiskit at $L = 8$ and 16 on `ibm_marrakesh` (QPU 100 s). Its pre-registered hypotheses were H1, accuracy above 0.5 in each cell, and H2, PSF-Zero's predictions closer to the device than Qiskit's.

cell	correct / points	H1 (Holm)	accuracy (twin predicted)	scale vs twin
PSF-Zero $L = 8$	109 / 150	confirmed	0.733 (0.771)	1.54
Qiskit $L = 8$	79 / 150	not confirmed	0.527 (0.739)	0.63
PSF-Zero $L = 16$	277 / 500	not confirmed	0.563 (0.624)	2.30
Qiskit $L = 16$	305 / 500	confirmed	0.625 (0.648)	1.00

Table 2: Device day (declared). "Scale" is the slope of the device's per-point z on the twin's (post hoc); the correlations are 0.970, 0.721, 0.969 and 0.964. H2 was not confirmed at either depth.

The twin had the right shape and the wrong scale (post hoc). Per point, the device's output followed the twin's closely (correlation 0.96–0.97 in three cells) but PSF-Zero's signal was 1.5–2.3 times larger on the device than the twin predicted, while Qiskit's $L = 16$ matched (1.00). On the device PSF-Zero kept more signal than Qiskit (λ 0.71 against 0.55 at $L = 16$), the reverse of the twin. The calibration had moved in the nine days since the snapshot (PSF-Zero's qubits: mean CZ error $0.0077 \to 0.0061$, median T_1 $144 \to 181\,\mu$s); scaled by the twin's own sensitivity, that drift explains about 1.2×, not 2.3×. The device also added an offset ($\delta = +0.10$ to $+0.28$) toward label $+1$.

The task needed no learning (post hoc). The device day's teacher has angle 0.3. Reading the *untrained* model on the same test points:

noiseless accuracy	$L = 16$ (500 points)	$L = 8$ (150 points)
untrained, $\theta = 0$	**0.934**	**0.940**
trained, PSF-Zero (item 31)	0.762	0.880
trained, Qiskit	0.730	0.787

Table 3: The device day's task, logical model only. Training on the twin made both arms' models worse than the start.

Both arms' training made the model worse than where it started. The day's comparison therefore measured which arm's training moved its model less far from a good start, and how each circuit read that model out — not which compiler trains the better classifier. The day's verdicts stand as recorded; what they measured is narrower than their question.

4 Confound 1: the teacher at the start

The effect is not specific to the device day. In a long-training run (declared; $L = 16$, teachers 11–13 at angle 0.3, 16,000 loops, checkpoints from 500), the untrained model scored 0.656, 0.904 and 0.962 noiseless on the three teachers. Both arms got worse with more training on every measure and every teacher: the zipper's shot-free stand-in accuracy fell from 0.87–0.94 at 500 loops to 0.71–0.74 at 16,000, item 31's from 0.63–0.68 to 0.53–0.59. The noiseless training accuracy stayed at 0.80–1.00 while the training loss rose: the outputs grow extreme. One reading, not tested: the loss $(1 - yz)/2$ rewards a large $|z|$ and does not penalise a confident wrong answer.

The declared rule of that run (noiseless test cross-entropy, Wilcoxon) found the zipper's trained model better in 3 of 3 teachers at 16,000 loops. That is a difference between trained models, but on such a task "better" means "degraded less": both arms were worse at 16,000 loops than at 500 on every measure. Separately, the zipper's large lead in stand-in accuracy on teachers 12 and 13 was mostly present at $\theta = 0$, where the two arms run the same logical model and differ only in how the circuit reads it out (Section 5).

Check: report the untrained model's accuracy on the test tier, per arm, before reading a training result.

5 Confound 2: readout offsets

Write the noisy readout as $z_{\text{noisy}} \approx \lambda z_{\text{noiseless}} + \delta$. Accuracy reads the sign, so an offset δ moves the decision threshold. On the twin, at $L = 16$ on the device day's model:

twin, $L = 16$	item 31	zipper	Qiskit
scale λ	0.268	0.183	0.496
offset δ	+0.046	+0.011	+0.087
δ/λ	0.17	0.06	0.18
fit residual (rms)	0.075	0.024	0.100

Table 4: Scale and offset fitted from the twin against the noiseless model on the 64 training inputs (no labels). The zipper's residual is about the trajectory noise of 1,024 trajectories (not checked with more): on the twin its noise acts as one uniform shrink.

The offset points toward $|0\rangle$ (label +1). For item 31, switching off relaxation in the twin lowers δ from 0.046 to 0.028: relaxation toward $|0\rangle$ is part of it, consistent with the non-unital bias studied in [7]; the rest is not attributed. No single noise source makes item 31's noise a uniform shrink.

On the stand-in the effect is large. At $\theta = 0$, where all arms run the same logical model, the share of +1 answers on balanced test points is (teachers 17, 18, 19 at angle 1.0): item 31 0.93, 0.92, 0.85; Qiskit 0.93, 0.94, 0.86; zipper 0.83, 0.78, 0.74 (Figure 1). On teacher 18 the untrained model is 0.640 accurate noiseless; read through Qiskit's circuit it is 0.504 and through item 31's 0.520, while the zipper's reads 0.624.

Two consequences. First, accuracy and loss can disagree for a structural reason: accuracy reads the sign and is sensitive to δ; cross-entropy reads the magnitude and rewards a large λ. An arm

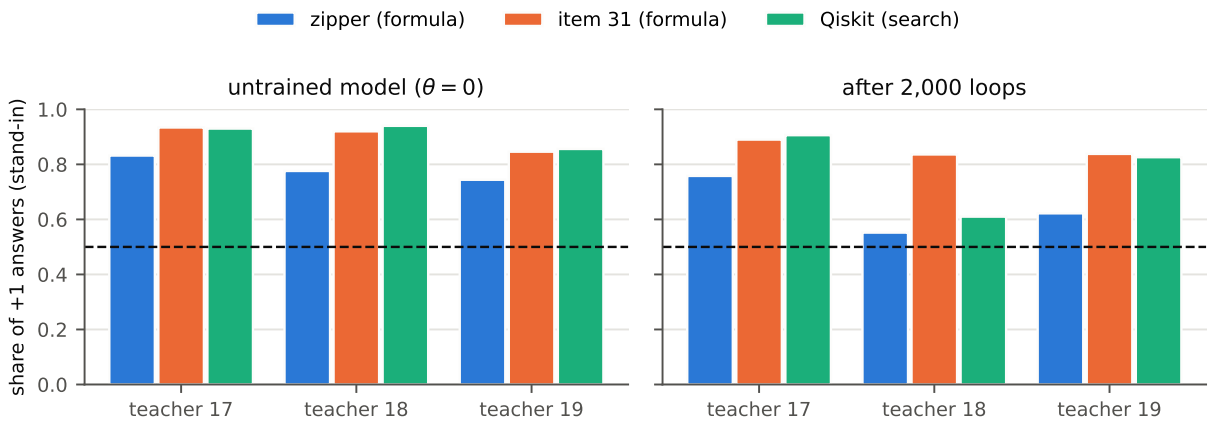


Figure 1: Share of +1 answers on the stand-in (shot-free), far teachers. Labels are balanced (dashed line, 0.5). Left: the untrained model, identical in all arms; right: after 2,000 loops.

with a small λ and no offset can win on accuracy and lose on loss, and the reverse. On the far teachers at 500 loops Qiskit had the lowest stand-in cross-entropy of the three arms on teacher 18 (0.585, with 250 shots) and an accuracy of 0.568, against the zipper's 0.635 and 0.711. Second, a gain measured as "accuracy after training minus accuracy at $\theta = 0$" does not cancel the offset when the offset saturates one class: item 31 on teacher 19 gains +0.068 noiseless but −0.020 on the stand-in by 2,000 loops.

Check: report the share of each answer next to accuracy, and an offset-free measure (AUC, or the trained model's noiseless accuracy).

6 Confound 3: unequal shot fragility

The zipper's output is shrunk more than item 31's on the twin, and the gap widens with depth (λ 0.375 against item 31's 0.398 at $L = 8$; 0.113 against 0.249 at $L = 20$; means over three teachers). To read the zipper's decisions with the noise of 250 shots of item 31's, the device would need 4.8× the shots at $L = 20$. In the declared data test ($L = 16$, 256 training points) the zipper's shot-free accuracy was 0.80–0.83 against item 31's 0.61–0.65; with the run's 250 shots it was 0.70–0.74 against 0.63–0.64. From shot-free to 250 shots at 2,000 loops on the far teachers the zipper lost 0.016, 0.017 and 0.046 and Qiskit, whose twin scale is the largest on the device day's model, −0.001, 0.044 and −0.007. Item 31 lost 0.001, −0.034 and −0.058: for an arm with a saturating offset, shot noise can *raise* accuracy by randomising some of the answers the offset had fixed.

Check: report shot-free and shot-based accuracy, and the output scale, per arm; fix the shot count in the declaration.

7 Declared tests on teachers far from the start

7.1 Data and early learning, zipper against item 31 (angle 0.3; declared)

Two earlier declared tests at angle 0.3 are summarised for completeness. In the data test (teachers 14–16, 2,000 loops) more training data was not the limit (256 against 64 points: WIN in 1 of 3 teachers for item 31, 0 of 3 for the zipper), and the zipper's raw accuracy lead survived it (3 of 3; 0.804/0.610, 0.830/0.644, 0.818/0.652; Holm $p \leq 3 \times 10^{-13}$). Post hoc on the long training, on

teacher 11 — the one teacher that needed learning — item 31 read the untrained model out better (δ removed: 0.718 against 0.636), and in 500 loops the zipper gained +0.242 against item 31's +0.018 (stand-in AUC 0.717 → 0.957 against 0.783 → 0.864). One teacher is a hypothesis, not a result; the next test was declared to check it.

7.2 Zipper against item 31 on far teachers (declared)

Teachers 17, 18, 19 at angle 1.0; untrained noiseless accuracy 0.574, 0.640, 0.498. Rule A: shot-free stand-in accuracy, zipper against item 31, McNemar, Holm over 6. Rule B: what training added, per point $g = \text{correct(trained)} - \text{correct}(\theta = 0)$ per arm, Wilcoxon on $g_{\text{zipper}} - g_{\text{item 31}}$.

	500 loops		2,000 loops	
teacher	A: zipper / item 31	B: gain zipper / item 31	A: zipper / item 31	B: gain zipper / item 31
17	0.584 / 0.546	+0.008 / +0.008	0.622 / 0.578	+0.046 / +0.040
18	**0.776 / 0.538**	**+0.152 / +0.018**	**0.780 / 0.636**	+0.156 / +0.116
19	0.576 / 0.554	+0.076 / +0.020	**0.658 / 0.514**	**+0.158 / −0.020**
verdict	A: no (1 of 3); B: no (1 of 3)		A: **yes** (2 of 3); B: no (1 of 3)	

Table 5: Far teachers, zipper against item 31 (stand-in, shot-free). Bold: WIN after Holm at $\alpha = 0.01$.

- **Vanity:** at 2,000 loops the zipper is the more accurate (2 of 3); on teacher 18 at 500 loops by +0.24.
- **Essential:** the declared "learns more" (B) is not supported at either budget. Item 31 answered +1 on 84–94 % of the balanced points at every budget, so part of A's verdict is its offset. B's premise — that the offset cancels in the gain — failed where the offset saturates (Section 5).
- **Post hoc:** the trained models themselves (noiseless accuracy, McNemar, Holm) differ in 1 of 3 teachers at 500 loops and 0 of 3 at 2,000. In noiseless gain the zipper learned the same on teacher 17 (+0.056 / +0.064), faster on 18 (+0.130 / +0.010 at 500 loops) and more on 19 (+0.142 / +0.068).

7.3 Formula against search: zipper against Qiskit on far teachers (declared)

Same plans; the Qiskit arm was trained and verified anew, the zipper's records reused (identical test labels, checked). Rule A as above; Rule B is the trained model's noiseless accuracy (declared to avoid B's flaw above). A device run of the comparison was to be declared only if the power of one McNemar test at 300 points, simulated from the stand-in with 250 shots, reached 0.8 in 2 of 3 teachers at one budget.

- **Vanity:** the zipper beats Qiskit on teacher 18 at 500 loops (0.776 against 0.554, Holm $p = 4 \times 10^{-14}$) with a third fewer CZ, a quarter of the duration and 4 fewer qubits.
- **Essential:** by the declared rules neither compiler gives the more accurate classifier, and what they learn cannot be told apart. A tie on one seed and three teachers is not a proof of equality. The device threshold was met only on teacher 18 at 500 loops (power 0.96); by the declaration, no device time was spent.

	500 loops		2,000 loops	
teacher	A: zipper / Qiskit	B: zipper / Qiskit	A: zipper / Qiskit	B: zipper / Qiskit
17	0.584 / 0.566	0.594 / 0.626	0.622 / 0.570	0.630 / 0.630
18	**0.776 / 0.554**	0.770 / 0.764	0.780 / 0.778	0.786 / 0.818
19	0.576 / 0.614	0.590 / 0.624	**0.658 / 0.582**	0.640 / 0.600
verdict	A: neither (1 WIN); B: neither (0)		A: neither (1 WIN); B: neither (0)	

Table 6: Far teachers, zipper (formula) against Qiskit (search). A: stand-in, shot-free; B: trained model, noiseless. Bold: zipper WIN after Holm; there was no LOSS.

- **Reported:** Qiskit's offset persists after training (+1 on 81–93 % of points) except on teacher 18 at 2,000 loops (0.61), where its accuracy rose to 0.778. The twin predicts the zipper's circuit better than Qiskit's (per-point rmse of $P(1)$, twin against stand-in: 0.008–0.014 against 0.016–0.055); at 500 loops Qiskit's stand-in cross-entropy fell outside the twin's 99 % interval on two teachers. Stand-in AUC at 2,000 loops: 0.762/0.771, 0.910/0.909, 0.739/0.730.
- **Cost:** the Qiskit arm took 4 h 51 min on an H100 against the declared 1–1.5 h; its trajectories evolve 14 active qubits, a state $2^{14}/2^{10} = 16$ times larger than the zipper's. This is a simulation cost; on hardware, by the billing model TOQB uses (0.036 s per circuit plus 0.27 ms and the circuit's duration per shot), 250 shots of a test point take about 0.12 s for Qiskit's circuit and 0.11 s for the zipper's.

Before the run, the workplace instance of the project wrote eight predictions for this comparison. Five held (Qiskit near chance at $\theta = 0$; AUC within 0.05; no consistent winner in what was learned; Qiskit less shot-fragile; teacher 17 hard for every arm), two held in part (Qiskit's +1 share at least 0.80: 8 of 9 cells; the twin at least 0.02 off on Qiskit's circuit: at $\theta = 0$, in 2 of 3 teachers at 500 loops, not at 2,000), and one failed: the zipper did not win raw accuracy at 2,000 loops in 2 of 3 teachers.

8 Discussion

What the formula buys. On this classifier at $L = 16$, formula placement without calibration was not distinguishable from calibration-aware search by either declared rule (one zipper WIN under rule A at each budget, no LOSS), with 336 instead of 506 CZ and 11.7 instead of 48.2 μs. How Qiskit's longer circuit keeps up is not established; one possible reason, untested, is its calibration-aware choice of qubits: on the device day's model its twin scale λ is 2.7 times the zipper's despite the longer circuit. The two routes reach classifiers the tests could not tell apart, by different trades. Where the formula should pull ahead is where duration and qubit count grow faster than good qubits can be found — deeper or wider circuits — and in reproducibility, since the formula's circuit does not change with the day's calibration. Neither is shown here.

What the zipper does and does not do. Against PSF-Zero's own earlier template, the zipper reads the model out with a much smaller offset, and on the twin its noise acts as one uniform shrink. The twin predicts the zipper's circuit about as well as item 31's, and better than Qiskit's. It did not, by the declared test, learn more. Its smaller λ costs shots, increasingly with depth.

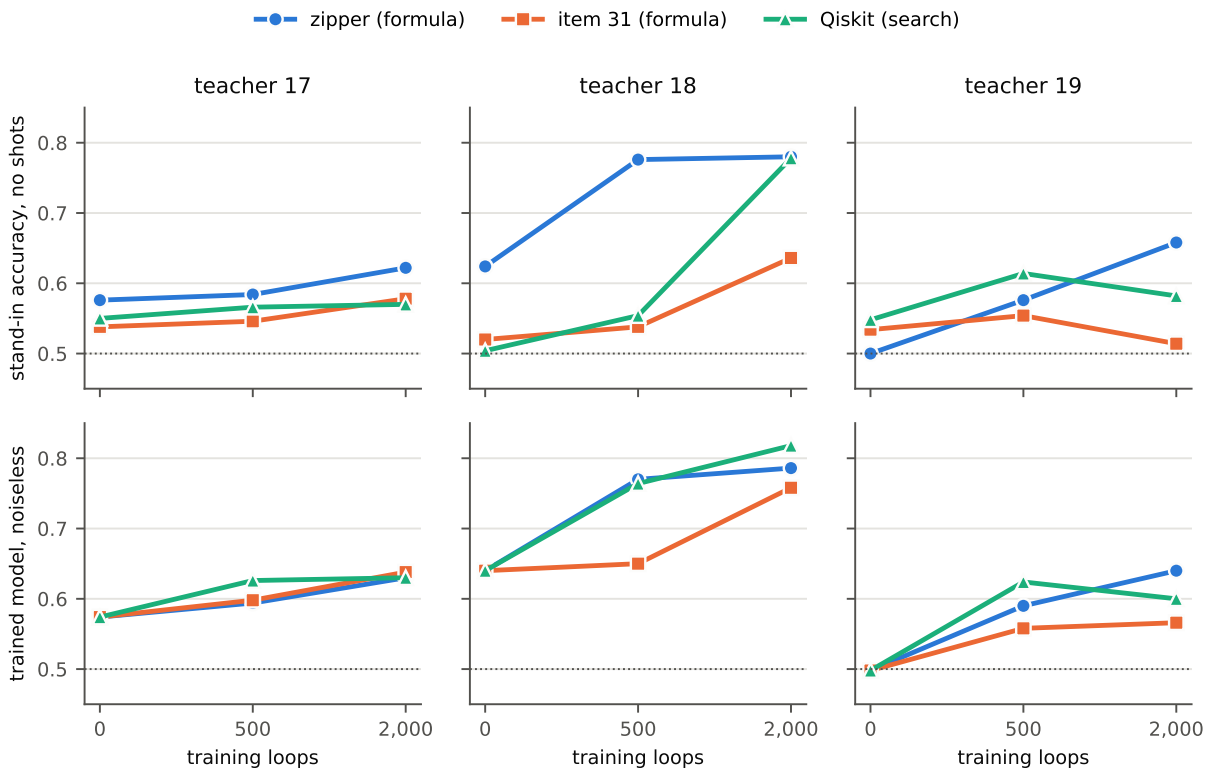


Figure 2: Far teachers: accuracy at $\theta = 0$, 500 and 2,000 loops for the three arms. Top: stand-in, shot-free (what the circuit delivers). Bottom: the trained model, noiseless (what was learned). At $\theta = 0$ the bottom row is one point: all arms start from the same model.

The confounds are general. None of the four effects is specific to PSF-Zero or Qiskit. Any comparison of compilers, devices or noise levels through a trained classifier can be won by a readout offset, a near-start task, a shot count that favours one output scale, or a device model off by a constant factor. Each was present in this project's earlier results, and each changed a conclusion.

9 Checklist for compiler comparisons through trained models

1. Report the untrained model ($\theta = 0$) per arm on the test tier, and place teachers where learning is needed.
2. Report the share of each answer next to accuracy; an arm far from the label prior has an offset.
3. Report an offset-free measure: the trained model's noiseless accuracy, or AUC.
4. Report shot-free and shot-based accuracy and the output scale per arm; fix the shot count in advance.
5. Measure the device model's scale against hardware (per-point regression) before trusting predicted absolute values.
6. Separate declared from post hoc results, and give the vanity and the essential number together.

10 Limitations

- One training seed; three teachers per test; $n = 10$; $L = 16$ for the declared far-teacher tests.
- All learning results are on simulators. The stand-in shares the twin's calibration base. The zipper has never run on hardware; the device day ran item 31 and Qiskit only.
- The post hoc analyses were guided by earlier results; their p-values are descriptive.
- The twin's scale was off by up to 2.3× on hardware for PSF-Zero; shot costs derived from the twin's λ are therefore uncertain on hardware.

11 Reproducibility

All runs, records and scores are in the TOQB test bench (https://github.com/TN-Holdings-LLC/TOQB), each result folder with the declaration it was scored against and, where the run produced one, the zip's SHA-256: device day `results/device-2026-10-22` (with `theta0_home`); long training `results/zip-long-2026-10-23`; data test `results/data-grad-2026-10-23`; depth sweep `results/zip-depth-2026-10-22`; noise sources `results/lam-ablate-2026-10-22`; far teachers `results/far-teacher-2026-10-23` (declared in `f5ed070`); zipper against Qiskit `results/far-qiskit-2026-10-23` (declared in `fe488cb`). The compilers are Qiskit 2.5.2 and PSF-Zero `9f0d44e` for the far-teacher tests (`6269c4f` for the device day, the depth sweep and the noise-source runs). The figures are computed from those records by `docs/papers/qml_confounds_figures.py`.

References

[1] TN Holdings, "PSF-Zero: An Analytic Two-Qubit Gate Synthesizer Combined with a Verified, Ordering-Aware Layout Search for Quantum Circuit Transpilation", Version 2 (2026). https://github.com/TN-Holdings-LLC/psf-zero

[2] A. Javadi-Abhari et al., "Quantum computing with Qiskit", arXiv:2405.08810 (2024). https://arxiv.org/abs/2405.08810

[3] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, J. I. Latorre, "Data re-uploading for a universal quantum classifier", Quantum 4, 226 (2020). https://quantum-journal.org/papers/q-2020-02-06-226/

[4] S. Wang et al., "Noise-induced barren plateaus in variational quantum algorithms", Nature Communications 12, 6961 (2021). https://arxiv.org/abs/2007.14384

[5] J. Vovrosh et al., "Simple mitigation of global depolarizing errors in quantum simulations", Physical Review E 104, 035309 (2021).

[6] M. Urbanek et al., "Mitigating depolarizing noise on quantum computers with noise-estimation circuits", Physical Review Letters 127, 270502 (2021).

[7] V.-Q.-M. Nguyen, T.-V. Truong, H.-L. Nguyen, T.-K. Le, "Zero- versus infinite-temperature damping in variational quantum circuits: feature scale, sampling cost, and frame gauge", arXiv:2610.01466 (2026). https://arxiv.org/abs/2610.01466

[8] M. Kashif, M. Shafique, "Late breaking results: hardware-aware compilation reshapes trainability in variational quantum circuits", arXiv:2604.16527 (2026). https://arxiv.org/abs/2604.16527

[9] M. Kashif, M. Shafique, "Beyond logical circuits: hardware-aware analysis of expressibility and trainability in variational quantum algorithms", arXiv:2605.25552 (2026). https://arxiv.org/abs/2605.25552

[10] J. Bowles, S. Ahmed, M. Schuld, "Better than classical? The subtle art of benchmarking quantum machine learning models", arXiv:2403.07059 (2024). https://arxiv.org/abs/2403.07059

[11] B. O'Gorman, W. J. Huggins, E. G. Rieffel, K. B. Whaley, "Generalized swap networks for near-term quantum computing", arXiv:1905.05118 (2019). https://arxiv.org/abs/1905.05118

[12] T. Parella-Dilmé, J. S. Kottmann, A. Acín, "Swap network augmented ansätze on arbitrary connectivity", arXiv:2507.23679. https://arxiv.org/abs/2507.23679